Talha Khan

Senior Game Developer - Software Engineer

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Professional Summary

Senior game developer and Unity/C# software engineer with 9+ years of mobile game development experience and 30+ published titles. Own the complete game engineering lifecycle, from prototyping and gameplay architecture to performance optimization, testing, LiveOps and Android and iOS delivery. Lead engineering teams, design free-to-play economies and develop mobile online services. Selected published games exceed 100M combined store-listing downloads.

Professional Experience

Ajax Studios

Technical Team Lead | Lahore, Pakistan | Aug 2019 - Present

Progressed through Junior, Senior and Development Lead roles.

- Grew the engineering team from one developer to eight through hiring, training and mentoring; led architecture, code review and delivery while remaining hands-on.
- Built Balanstein from scratch in Unity and own its engineering through active, unreleased production: a physics-based 2.5D arcade game with free-to-play economies, progression and adaptive difficulty.
- Designed modular Unity/C# systems using SOLID principles, dependency injection, state machines and event-driven interfaces; documented 47 architecture decisions for the online-services layer.
- Engineered deterministic physics replay for the recorded world, using a fixed-step clock, versioned compressed binary data, mid-act snapshots and coordinated timelines.
- Developed custom level-authoring tools supporting approximately 800 levels. The Social Mode tooling includes 18 custom inspectors, six editor windows and four property drawers for recording, baking, validation and hierarchy automation.
- Implemented Unity Gaming Services authentication for anonymous sign-in, Unity Player Accounts, Facebook and Google Play Games, including account linking and shared conflict handling.
- Built versioned cloud saves with local caching, optimistic concurrency and event-driven recovery. In a controlled batching test, 28 rapid state changes produced one cloud write after an explicit flush.
- Built and tested Local, Weekly and Global leaderboards with reusable, virtualized paging; implemented two-stage connectivity checks to identify captive portals.
- Delivered animated UI across mobile aspect ratios and 28 languages, including Arabic right-to-left layouts; integrated VFX and collaborated with technical artists on character animation.
- Profiled CPU/GPU frame time, memory, garbage collection and rendering against a 30-60 FPS target on 3 GB RAM devices to guide optimization.
- Integrated monetization and LiveOps across studio projects using AppLovin MAX, AdMob, Unity Ads, in-app purchases, AppsFlyer, Adjust, analytics, Remote Config and local/push notifications; modeled service costs for infrastructure planning.
- Delivered Android/iOS builds and store releases across studio projects, coordinating internal, alpha and closed testing with QA, automated Unity tests and CI/CD. Verified Balanstein authentication, cloud-save and recovery flows on Android devices.
- Introduced reviewed AI-assisted CLI and MCP workflows for implementation and developer tooling, retaining engineering judgment and code review.

ZEKAB International Limited

Game Developer | Rawalpindi, Pakistan | May 2018 - Jun 2019

- Shipped 16 Unity titles to Google Play in 13 months from approximately 20 prototypes, owning gameplay, progression, in-app purchases, ads, leaderboards and achievements through release.
- Helped establish the studio's Unity workflows and mentored junior developers and interns.

Desire Studios

Game Developer | Rawalpindi, Pakistan | Jul 2017 - Apr 2018

Promoted from Intern.

- Designed, developed and published 18 mobile titles across hyper-casual, simulation and puzzle genres, owning concept-to-deployment engineering and device optimization with designers and artists.
- Earned five consecutive Employee of the Month awards for creative game ideas and the lowest crash rates in personally developed titles.
- Refactored and optimized 23 additional studio games to reduce crashes and fatal errors.

Technical Skills

Unity & Gameplay: Unity 2D/3D, C#, gameplay programming, ScriptableObjects, custom controllers, Input System, Enhanced Touch, Blend Trees, object pooling, Unity Splines, Unity Mathematics, save/load, Netcode.

Software Engineering: Object-oriented programming (OOP), SOLID, clean architecture, design patterns, data structures and algorithms, async/await, dependency injection, cancellation, refactoring, code review.

Graphics & Animation: URP, HDRP, Shader Graph, HLSL shaders, particle systems, post-processing, VFX, character animation, technical-artist collaboration.

UI & Localization: uGUI, TextMeshPro, pixel-perfect and animated UI, Canvas/Sprite Atlas optimization, aspect ratios, localization, Arabic RTL, virtualized lists.

Profiling & Diagnostics: Unity Profiler, CPU/GPU and device profiling, frame time, memory management, garbage collection, draw-call reduction, Android logcat, device debugging.

Testing & QA: Unity Test Framework, automated tests, internal/alpha/closed testing, cross-platform and device validation, bug tracking, pre-release QA.

LiveOps & Backend: Unity Gaming Services (UGS) Authentication, Cloud Save, Cloud Code, Leaderboards, Analytics, Remote Config; Firebase Authentication, Firestore, Cloud Messaging (FCM), Analytics, Remote Config; REST APIs, Unity Mobile Notifications.

F2P & Game Economies: Free-to-play (F2P) economy design, resource sinks, progression, retention, replayability, adaptive difficulty, game business models, monetization analytics, A/B testing.

Monetization & Attribution: AppLovin MAX, AdMob, Unity Ads, ad mediation, SDK integration, in-app purchases (IAP), AppsFlyer, Adjust.

Release Pipeline & DevOps: Android/iOS, Google Play/App Store, IL2CPP, ARM64, Xcode, Android Studio, continuous integration and delivery (CI/CD), Unity Cloud Build, Git, Addressables, Asset Bundles, Unity Editor tooling, IMGUI, UI Toolkit.

Supporting Languages & AI: C++, Python, Java, JavaScript; reviewed AI-assisted CLI and Model Context Protocol (MCP) workflows using Claude Code, Unity MCP, Cursor and Codex.

Leadership & Production: Hiring, mentoring, technical guidance, Agile/sprint planning, code review, collaboration with designers, artists and QA.

Genres & Game Models: Vehicle simulation, physics arcade, hyper-casual, puzzle, action/shooter, free-to-play, live service.

Education

Bachelor of Science (Hons) in Software Engineering

Riphah International University | Islamabad, Pakistan | 2013 - 2017